Project Activity Log

Learner Name	Louis Vidal	**Learner number**	
Centre Name	Highgate School	**Centre Number**	12610
Unit Name	Artefact	**Unit number**	P304
Teacher Assessor			
Proposed project title			

This form should be used to record the process of your project and be submitted as evidence with the final piece of work.
You may want to discuss:

- what you have done (eg, from one week to the next)
- if you are working in a group, what discussions you have had
- any changes that you have or will need to make to your plans
- what resources you have found or hope to find
- what problems you are encountering and how you are solving them
- what you are going to do next

Date	Comments
24/11/24	First EPQ planning session. I've been interested in AI and wanted a project that could have a real-world impact. I started looking into how machine learning is being used in sustainability and environmental science to find a focus for my project.
01/12/24	I want to find a way in which AI could help solve some common ecological issues. I am currently looking at computer vision and image processing and looking at how those technologies can be used to improve current solutions to some ecological issues. Some issues that I am considering is waste classification and precision farming where detect plant diseases from images.
10/12/24	Following further research and after looking at existing implementation of technology to protect the environment, I have selected automated rubbish classification as my project focus due to its potential for a more significant environmental and societal impact. This approach can be utilized to replace the current manual labour in waste facilities, a task that is often tedious, unhygienic, and inefficient.
14/12/24	I did some digging into how other are tackling automated waste sorting. It looks like the accepted way of solving image processing tasks is by using CNNs because they're great with images. However, I did develop a theory, most rubbish has very distinct colours like brown cardboard versus green glass, so maybe a simpler MLP could actually do the job. I want to test this out by building both types starting with the MLP. I have submitted my Project

	prosposal form earlier this week.
22/12/24	Today, I found a dataset on Kaggle for garbage classification. Thought about making my own dataset but it would be very time-consuming. This dataset has a good image with 6 types of rubbish which should be enough for this project. In the future, I can find better datasets on Kaggle as it offers great resources for free.
25/12/24	I started looking into the maths for implementing the multi-layer perceptron. Currently, I am looking at different articles to see how to go about implementing an MLP. For this project I need basic calculus and linear algebra, so I am looking at different YouTube videos and documentation to set the foundations.
05/01/25	I researched how image data is typically formatted for neural networks and looked into techniques like normalization for my dataset.
12/01/25	I researched into the architectural limitations of MLPs for image data. They need to flatten each image into a one-dimensional vector meaning the model will struggle with patterns, edges and shape. This differs from my initial hypothesis. Reading a passage of chapter 9 of Goodfellow's Deep Learning book really clarified my perspective. In summary, the MLP are inefficient for high-dimensional data like images. As to learn, in this space, it needs an enormous amount of training examples and therefore has a very poor improvement rate compared to CNNs. Furthermore, CNN don't care if an image of bottle for example is in the corner, upside down etc, while this makes a huge difference in MLPs (this is called translation invariance).
26/01/25	Researched activation functions in MLPs, including sigmoid, tanh, and ReLU, and noted their advantages and disadvantages in image classification tasks. Found that ReLU is often better for deeper models due to avoiding vanishing gradients.
02/02/25	I studied basic MLP architectures and compared them to CNNs to understand their strengths and weaknesses for image classification. I watched videos like the 3Blue1Brown Neural Network series? Additionally, I have explored scientific papers on google scholar such as Backpropagation Learning Algorithm (Rumelhart et al., 1986) and Krizhevsky's 2012 paper on Alex Net. I couldn't get to much related information out of the original 1986 paper but the 2012 paper was much more useful. I had some additional programming issues with processing the data as my current code logic doesn't work with the indexed text file that handles the classification of each individual datapoint. Furthermore, I have wrongly stored the pixel values for each image, so instead I have stored each pixel value as cell in a csv file where each row represented a separate image, this is very RAM costly and is not scalable.
23/02/25	After some research I have settled on using a CNN to process the dataset as it consistently leads to better results due to its superior ability to recognise spatial features in images. This will be useful as the images I am processing have some more nuanced differences which theoretically will be harder to find. Going forward I will still implement the multi- layer perceptron so that I compare both models on different metrics.
09/03/25	I outlined the initial model structure, deciding on the number of layers and neurons. I also looked at activation functions to determine which might work best. When implementing the model, I will test different layer sizes and model layer depths to compare differences.
16/03/25	I set up my coding environment, installing necessary libraries like PyTorch, Torchvision, NumPy, Pandas, and Scikit-learn. I also installed the relevant CUDA drivers for my GPU so that I can locally train my model.
23/03/25	I wrote basic preprocessing scripts, including resizing and normalizing images. I had to get familiar with pandas to manage my dataset. For my dataset, I have settled with a dataset on Kaggle that has classified the images of rubbish into 6 categories (https://www.kaggle.com/datasets/asdasdasasdas/garbage-classification?select=Garbage+classification). In addition, I have created a GitHub repository to store all my relevant files in the future.
30/03/25	I have started to read on how to implement CNNs in python. I have both explored tenser flow and PyTorch documentation as well as tutorials on websites like medium.

06/04/25	I had some additional programming issues with processing the data as my current code logic doesn't work with the indexed text file that handles the classification of each individual datapoint. Furthermore, I have wrongly stored the pixel values for each image, so instead I have stored each pixel value as cell in a csv file where each row represented a separate image.
13/04/25	Started writing the code for implementing an MLP in PyTorch. I'm also building a simpler one from scratch in NumPy using a tutorial by El Caiseri to solidify my understanding of the underlying maths of backpropagation.
20/04/25	I am still working on the code for implementing the MLP while also reviewing my knowledge on MLPs on my notes.
27/04/25	Solidified mathematical knowledge of forward propagation and linear regression.
04/05/25	I have some issues running the code as my GPU is not recognised. This is an issue as training a model on CPU is much slower. I have tried reinstalling my CUDA drivers and resetting my virtual environments. I have been struggling on this problem for a while now. I am running this on a newly built pc and I am not too familiar with connecting the drivers to my pytorch implementation. I have completed the implementation of the MLP. Training results are poor around 32%. This aligns with the theoretical limitations I identified earlier.
11/05/25	Finished writing my CNN model using Pytorch. My issue regarding has been resolved after a lot of trial and error. I had the wrong Cudnn version installed. Next, week I am aiming to compare both models on accuracy.
18/05/25	This week, I conducted final training runs to ensure results are consistent. The training is relatively quick considering my GPU is now compatible. The MLP from scratch implementation has a very low accuracy, close to random around 24%. The more complex MLP I built with pytorch implementation plateaus around 30 to 33% despite changing the model size. This indicated the model's inability to learn more complex pattern and features. On the other hand, the CNN model has achieved much higher accuracy of around 71%. This clearly shows that the CNN is superior when it comes to this task.
25/05/25	I investigated how models are scored and compared. What I have found was that in general, accuracy can give an overview of how the program is doing. However, this is only scratching the surface in comparing models. A common way to score a model is to use the f1 score, this is a harmonic mean between the recall and the precision. A harmonic mean is used so that if one of recall or precision is low the F1 is score is low even though the other one might be high. The model is penalised if precision or recall is low. In addition, I am using sklearn to generate confusion matrices, which are very useful in modelling the models' faults. I found the explanation of these concepts on Papers with Code and towardsdatascience.com very useful. The results clearly show the CNN excels at classifying distinct materials like cardboard and paper but struggles more with visually similar classes like glass and plastic. The MLP's performance is consistently weak across all categories, validating the decision to transition to a CNN architecture.
01/06/25	I started writing the report. I began with the introduction and the background research. I'm focusing on explaining why I chose to explore both models and why the CNN was the clear winner.
08/06/25	This week was all about the development section. I wrote about the data preprocessing and how I set up the training environment. I also explained how I implemented the models in PyTorch and dealt with the GPU driver issue I had earlier.
15/06/25	Wrote the evaluation and reflection section. I included the confusion matrix and the classification report here and talked about what the results meant. I also reflected on the limitations of my project and what I learned from it, especially about the importance of using the right model for the job.
22/06/25	I worked on the conclusion and the future improvements part of the report. It was cool to think about how I could take this project further, like using a bigger dataset or implementing transfer learning with a pre-trained model.
01/07/25	Completed a full proofread of the first draft. Focused on correcting grammatical errors and improving the overall flow. I am conscious that the report currently focuses heavily on the technical outcome. I need to make sure my final drafts accurately depict the journey that I went through for completing this project, for example how I switched from MLP to CNN.
10/09/25	I finalized all my citations and the bibliography. I also made sure all my figures and tables were properly labelled, and I cleaned up some of the sentences.

12/09/25	Implemented my supervisor's feedback for the final draft. I added more reflective commentary on why the MLP's poor performance was a valuable learning step, not just a setback.
17/09/25	This is week I am starting to prepare for my presentation next week. To suit the audience, I need to make my presentation more about the journey and less about the technical details.

Reflection on Timing of progress

Looking back at my original project proposal form, I managed to follow my planned timeline quite closely. As planned, my research phase was completed around February 2025, and the model design and training were finished by May 2025. However, I did experience short delays, due to issues with GPU implementation and debugging, which slightly extended the write up to the autumn term. Overall, my time was managed well with the original planning leaving some room for delays such as the one I experienced.